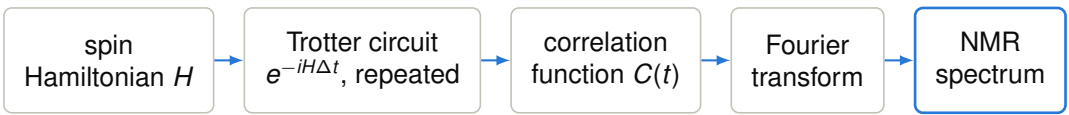
Matching molecules to NMR spectra with Trotterized time evolution

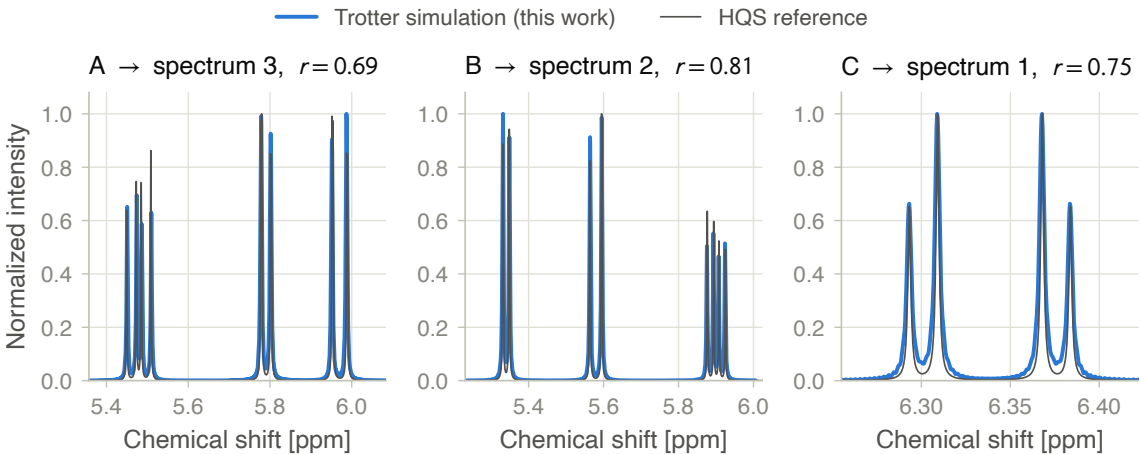
Jonas Merkinger Pascal Fries Michail Sokolov Sajedeh Shahbazi Tobias Fellner

Qiskit Fall Fest 2026 · intermediate track

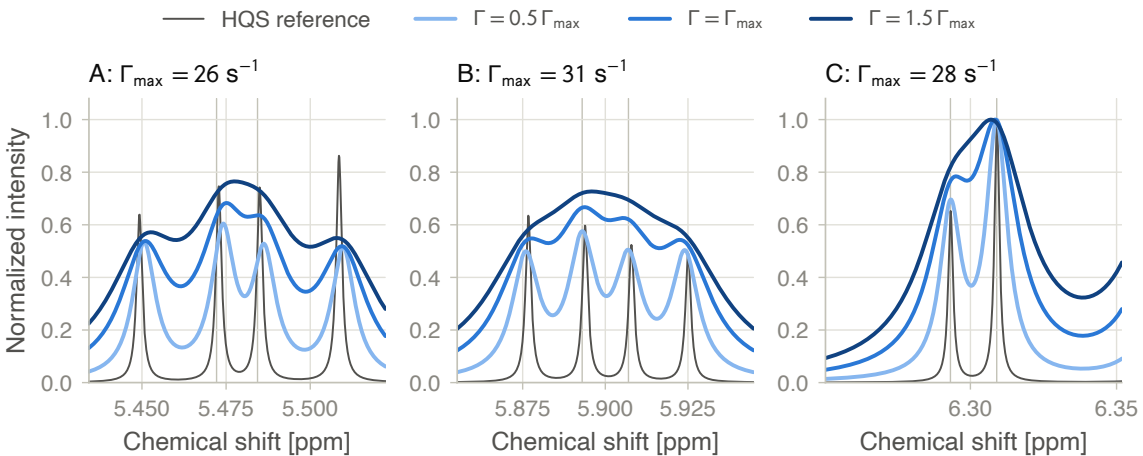
From spin Hamiltonian to spectrum



Simulation identifies every molecule



Lines stay resolved up to $\Gamma \approx 26\,\mathrm{s}^{-1}$



Spectra on IBM Heron hardware

60 Trotter steps

